

Falk Woldmann Lu

Senior software engineer specializing in Rust with strong experience in distributed systems, cloud infrastructure, DevOps, and event-driven architectures. University lecturer specializing in Rust programming, as well as an active open-source contributor and co-maintainer within the Rust ecosystem.

Professional Experience

Otto GmbH & Co. KGaA

Senior Software Developer 2025/03 – Present

- Technical lead for the adoption of Rust at OTTO, including mentoring teams, supporting the first production Rust services, and establishing internal knowledge-sharing formats.
- Key contributor to OTTO’s cloud efficiency initiatives, focused on identifying and implementing measures to reduce infrastructure and operational costs across multiple services and platforms.
 - Drove the design and implementation of a company-wide initiative that reduced annual cloud data transfer costs by more than \$500,000 through IPv6 and dual-stack networking strategies. Additional details are outlined in this [technical blog post](#).
- Development of scalable distributed systems and micro-frontends with a strong focus on reliability, observability, and operational excellence, leveraging agentic engineering tools to improve development efficiency in agile cross-functional product teams.
- Mentoring and supporting dual-study students and apprentices.
 - Scientific and business supervision of the best bachelor’s thesis in the 2026 Applied Informatics cohort at NORDAKADEMIE on evaluating Rust design patterns in production systems: [Public Thesis Repository](#).

Software Developer 2022/03 – 2025/03

Junior Software Developer 2021/03 – 2022/03

Dual Study Program in Business Informatics 2017/08 – 2021/03

NORDAKADEMIE University of Applied Sciences

Adjunct Lecturer & Academic Examiner 2025/10 – Present

- Lecturer for module *A215 Cloud and Distributed Systems* (45 semester hours, Q3 2027)
- Co-lecturer for the seminar *Introduction to the Rust Programming Language*
- Academic examiner and thesis evaluator for academic research papers and bachelor’s theses

Education

NORDAKADEMIE University of Applied Sciences

Bachelor of Science (Business Informatics), Final Grade: 1.7 2017/10 – 2021/03

Skills

Programming Languages: Rust, Python, Kotlin, Java	Architecture: Event Sourcing, Domain-Driven Design, Micro-frontends, REST APIs, distributed systems, event-driven architectures
Cloud & Infrastructure: AWS, Kubernetes, Docker, Terraform, GitHub Actions	Practices: TDD, Pair Programming, Scrum, Kanban, Disaster Recovery
Messaging: Kafka, SNS/SQS	Frontend: TypeScript, JavaScript, HTML, CSS
Databases: PostgreSQL, DynamoDB, MongoDB, SQLite	

Languages

German	English	Mandarin
Native	Full professional proficiency	Currently learning (A1.2)

Projects & Research

Academic Publication

- Leon Heuer, Falk Woldmann Lu, Jan Haase (2026). *Functional State Machines in Rust: Typestate and Newtype Patterns (Experience Report)*. In: Proceedings of the 4th ACM SIGPLAN International Workshop on Functional Software Architecture (FUNARCH 2026). DOI: 10.1145/3830438.3830958.
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Selected Writing

- Falk Woldmann, Nico Schmidt, René Hottendorf (2026). *Rewriting OTTO in Rust: oxidizing our codebase, one service at a time*. OTTO Techblog.
 - COST, Falk Woldmann (2024). *How we used a simple trick to save USD 500,000 in data transfer costs*. OTTO Techblog.
 - Falk Woldmann, Kay Hannay (2024). *Recap of the EuroRust 2023 conference in Brussels*. OTTO Techblog.
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Open Source

- Co-maintainer of [httpmock](#), a widely used HTTP mocking library for Rust with more than 600 GitHub stars and 20+ million downloads on [crates.io](#).
 - Contributions to various Rust ecosystem projects, including Tokio, Axum, Tower, Reqwest, and AWS SDK & Lambda libraries.
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Conferences & Community

- Speaker at bit summit (2026) with Maximillian Hausner: Talk on *NAT Gateway to Net Savings: How OTTO saved \$500k with IPv6*.
 - Speaker at CloudLand Conference (2026) with Jan Maximillian Hausner: Talk on *NAT Gateway to Net Savings: How OTTO saved \$500k with IPv6*.
 - Speaker at [AWS User Group Hamburg](#) (2025/05): Talk on *How to use a simple trick to save USD 500,000 in data transfer costs* with Michael Weidemann & Frederic Düwer, hosted at OTTO.
 - Organizer of AWS User Group Hamburg meetups.
 - Organizer of Rust User Group Hamburg meetups.
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